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Mini review

Effects of landscape heterogeneity on bird communities in temperate, boreal, and montane forests – a review

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Bird populations in Europe and North America have been strongly declining for the last 40 years. As mobile organisms, bird species are sensitive to landscape patterns; therefore, landscape effects on bird communities need to be understood to set relevant conservation measures. However, forest bird communities have received much less attention than their counterparts in agricultural landscapes in this respect. In this systematic review, we investigated the effects of landscape heterogeneity on bird communities in forest contexts by searching for empirical studies conducted in the boreal, montane, and temperate biomes. We found 45 articles from which we extracted 1272 single results (i.e. tested relationships between landscape and biodiversity metrics). We found that most of the articles studied local alpha-diversity (78%; 22% for landscape gamma-diversity) during the breeding season (87%). In contrast, most of the significant results were related to bird gamma-diversity, indicating a positive effect of landscape heterogeneity. Most of the tested landscape metrics were related to compositional heterogeneity (74%). Overall, habitat amounts had a large positive effect on their related communities (i.e. forest amount on forest bird species). However, forest composition variables (e.g. broadleaf forest proportion) mostly led to non-significant effects. While we found antagonistic results depending on bird habitat preference, groups based on migration strategy were largely unresponsive to landscape variables. However, we failed to find a consistent scale of effect across studies. We discuss the potential landscape mechanisms at play, such as niche partitioning, edge effects, and habitat complementation. We recommend better acknowledging forest ecosystem complexity and variability in future forest landscape studies and better recognition of bird habitat requirements beyond the breeding season (including overwintering sites and migration stopovers).

Keywords: avian, edge effect, habitat amount, landscape composition, landscape configuration

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Introduction

Global changes caused by human activities threaten biodiversity, with an estimated one million species at risk of extinction globally (IPBES 2019). Bird species are no exception, and long-term monitoring programs in Europe and North America have shown strong declines in their abundance and changes in their spatial distributions (IPBES 2019, Rosenberg et al. 2019, Rigal et al. 2023). Such declines are particularly severe and well-documented for farmland birds (Rosenberg et al. 2019, Rigal et al. 2023). Meanwhile, forest and woodland bird species have remained relatively stable overall, at least in Europe, during the past three to four decades (Gregory et al. 2019). However, forest bird species sharing specific biological traits or distributed in the boreal zone have also experienced important declines and range changes (Cours et al. 2025a).

Both land use and climate change affect forest bird communities and need to be considered in their conservation (Northrup et al. 2019, Rigal et al. 2023). Forestry degrades forest habitat at large scales (e.g. loss of large and old trees and deadwood), depriving forest birds of nesting sites (e.g. cavities) and food sources (e.g. saproxylic organisms; Lehikoinen and Virkkala 2018, Betts et al. 2022). Meanwhile, global warming forces range shifts for many bird species, including forest species (La Sorte and Thompson III 2007, Marjakangas et al. 2023, Cours et al. 2025a). For example, boreal forest bird species associated with mixed forests have shown a major decrease in abundance and range, potentially due to a bottleneck effect between forest management intensity and climate change (Cours et al. 2025a). These responses to global changes occur at the landscape scale, as birds are mobile organisms that often exploit multiple habitats. Landscape patterns shape bird communities by influencing dispersal and space occupancy, affecting their taxonomic, functional, and phylogenetic diversity (Klingbeil and Willig 2016). Beyond habitat loss, fragmentation can hinder species movement, limiting climate-driven range shifts – especially for species with limited dispersal capacity (Jarzyna et al. 2015, Marjakangas et al. 2023, Weeks et al. 2023). However, habitat quality can mitigate climate change effects. For instance, preserving old-growth forests within protected areas may not only offer refuge habitats for old-growth forest specialists (Barbaro et al. 2023, Santangeli et al. 2023), but also act as sanctuaries for species associated with cool temperatures (cold-dwellers; Lehikoinen et al. 2021).

Although landscape heterogeneity seems to play an important role in the conservation of forest bird communities, it has been less studied than in agricultural settings (Johnson et al. 2011, Sirami et al. 2019). Forests encompass diverse habitats, from broadleaf to coniferous forests, and from early successional stages to old-growth forests. Consequently, forested landscapes form mosaics of patches with varying tree species composition, developmental stages, and structures, shaping bird communities through multiple landscape-level mechanisms (Tews et al. 2004, Tscharntke et al. 2012, Duflot et al. 2022, Tonetti et al. 2023). This is especially true in forest production landscapes, where management actions create homogeneous patches and sharp edges between them, in contrast to most natural disturbance regimes (Kuuluvainen and Gauthier 2018). Such patterns are particularly visible in areas dominated by even-aged forestry (rotation forestry with clearcutting).

Landscape composition (i.e. habitat type, amount, and diversity) determines the quality, quantity, and diversity of niches available in a landscape, while landscape configuration (i.e. the spatial distribution of habitats) influences species composition via, for example, dispersal or edge effects (Duelli 1997, Fahrig et al. 2011, Tonetti et al. 2023). The availability of high-quality habitats, such as old-growth forests, supports specialised bird communities (Jansson and Andrén 2003). In addition, habitat diversity can enhance overall landscape-level bird diversity by accommodating species with varying biological traits (Barbaro et al. 2005, Culbert et al. 2013, Klingbeil and Willig 2016). Edges between habitats (e.g. of different tree species composition or successional stages) can benefit species requiring multiple resources at the landscape scale for different purposes (e.g. foraging vs nesting; Barbaro et al. 2007) or at different seasons (Yabuhara et al. 2019), illustrating habitat complementation (Dunning et al. 1992).

Using a systematic approach, we extracted results from empirical studies across temperate, montane, and boreal biomes that tested the effects of landscape heterogeneity on bird communities in forests. We particularly examined how these relationships have been studied and their key findings. We looked at which landscape metrics were used to describe habitat composition and configuration, and those driving the strongest biodiversity responses. Additionally, we documented the scales at which biodiversity was measured (local alpha- vs landscape gamma-diversity) and the spatial scales considered for landscape metrics. Our aim was to estimate the scale of effect for bird communities in forests, that is, the landscape scale to which a community is the most responsive, which is critical for defining appropriate conservation strategies (Jackson and Fahrig 2015, Monroe et al. 2022).

Material and methods

Literature search

We systematically reviewed studies examining the effects of landscape heterogeneity on bird communities found in forest ecosystems, within boreal, temperate, and montane biomes. Articles were retrieved from the Web of Science Core collection (October 2023) and supplemented with additional entries from Google Scholar (June 2024). We searched the selected terms within the titles, abstracts, and keywords in Web of Science, while Google Scholar applied the search string across the entire text. The review was not restricted by publication date. To refine our selection, we used specific terms (e.g. '*species diversity*' instead of '*diversity*', '*landscape composition*' instead of '*landscape*') to exclude, for instance, studies that referenced landscape context without explicitly analysing landscape heterogeneity effects. This approach

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ensured a more relevant dataset aligned with our inclusion criteria. The Web of Science search was made using the University of Jyväskylä's institutional access. The same search string was used for both the Web of Science and for Google Scholar:

('bird*' OR 'avian' OR 'avifauna') AND ('biodiv*' OR 'species diversity' OR 'species richness') AND (forest* OR 'woodland*' OR 'wood land*') AND ('boreal*' OR 'temperate*' OR 'montane' OR 'mountain') AND ('landscape configuration*' OR 'landscape composition*' OR 'landscape structur*' OR 'landscape heterogen*' OR 'landscape homogen*' OR 'landscape scale*' OR 'landscape complex*') NOT (Urban OR tropical)

The Web of Science search yielded 160 articles, to which we added 21 articles from Google Scholar, leading to a total of 181 articles considered in this review.

Inclusion criteria

We scrutinized all 181 articles and excluded those conducted outside the target biomes (i.e. temperate, montane and boreal) or not studying birds. Only studies that statistically assessed the effects of landscape heterogeneity on bird communities were included. We removed articles where landscapes were not described quantitatively, e.g. described as background context. Additionally, we only retained studies using direct biodiversity metrics (e.g. taxonomic, phylogenetic and functional diversity, abundance, or species composition change), excluding those focused on single species or ecological processes such as seed dispersion. Only field-based empirical studies were considered, leading to the exclusion of simulations, reviews, and meta-analyses. Finally, we included studies where observations were conducted at least partially in forested areas. After this selection process, data were extracted from 45 articles (Fig. 1).

Data extraction

First, we extracted general study information, i.e. publication year, location (country, geographic coordinates, and biome), main landscape features, bird sampling season, location of observation/sampling unit (forest-only, predominantly or partially in forest), scale of biodiversity measurement (i.e. alpha, beta, or gamma), sample size, number of tested landscape size(s) (usually buffers), and the minimum and maximum size of tested landscape scales.

Second, we extracted all tested relationships between the landscape and bird biodiversity metrics, including both

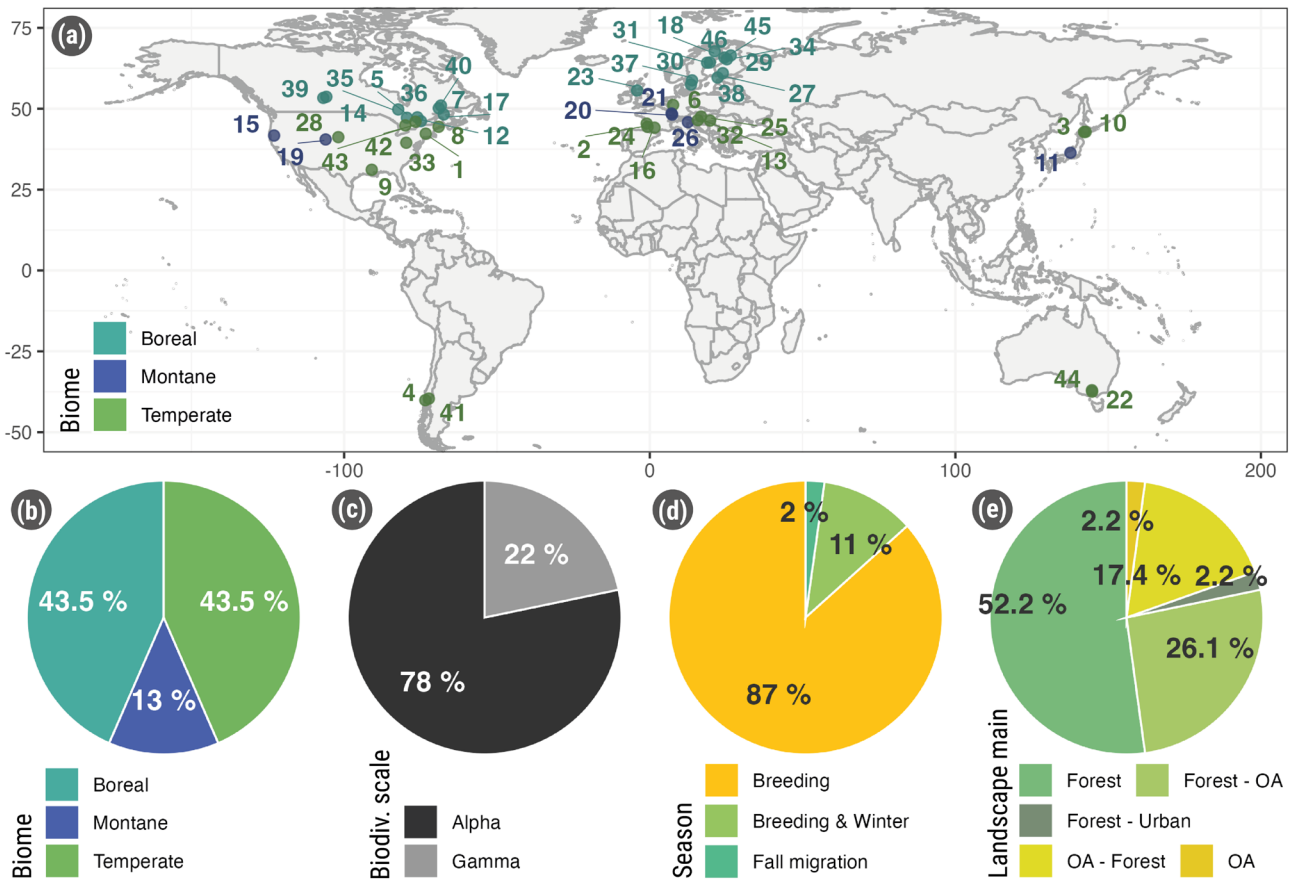

Figure 1. (a) Map of the 45 studies selected for qualitative and quantitative analyses (colour based on biome; Supporting information to see related studies). Percentage distribution of articles based on (b) biome, (c) the scale of biodiversity measurement (i.e. alpha- local diversity or gamma- landscape diversity), (d) the sampling season(s), and (e) the main land cover in the landscapes ('OA' = 'open areas').

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significant and non-significant results. For significant relationships, we recorded the direction of the effect (positive or negative) and the shape of the relationship if available (e.g. hump-shape), or the assemblage change when applicable. In studies incorporating local variables alongside landscape metrics, we also recorded their name and effects. To facilitate comparisons, we categorized landscape metrics into two hierarchical levels (Supporting information). First, similar metrics used under different names were merged (e.g. 'deciduous wood cover', 'amount deciduous forest', '%broadleaf forest' were combined as 'amount broadleaf forest'). Second, metrics were classified based on heterogeneity component (composition or configuration) and focal habitat. For example, general habitat composition (e.g. Shannon diversity of land cover) and forest composition (e.g. amount of broadleaf forest) were treated separately. Hereafter, we use 'amount' to refer to metrics quantifying specific habitat in the landscape, typically measured as proportional area. Finally, all metrics were categorized by heterogeneity component: composition, configuration, or mixed (e.g. 'Amount of forest core' which incorporates both).

We extracted the specific biodiversity metrics (e.g. species richness) and biodiversity type (e.g. taxonomic diversity). When the entire bird community was studied, results were categorized as 'All spp.'. For studies analysing species subgroups based on biological traits, we classified results accordingly (e.g. migration status: migrant vs resident). Given our focus on bird communities in forests, we distinguished species by habitat preference (forest vs open areas). For example, cavity nesters were classified as forest species, while early succession bird species were classified as open areas species (see the Supporting information for details). For each single result, we also recorded the scale of biodiversity measurement (i.e. local alpha or landscape-scale gamma diversity) and, for gamma-diversity, the spatial scale of the measurement.

To assess the landscape scale of effect on bird communities in forest, we analysed a subset of studies comparing the strength of landscape metrics across multiple spatial scales around the sampling unit. The scale of effect was defined as the spatial scale yielding the strongest (magnitude), the most significant (p-value), or the best fitting (AIC or R^2) relationship between a landscape and a biodiversity metric.

Results

General results

Of the 45 selected articles, 87% were conducted in boreal and in temperate biomes (equally distributed in both), and 13% in the montane biome (Fig. 1b). Studies were evenly distributed between Europe and North America, with a few from the Southern Hemisphere (two from Chile, two from Australia), and three from Japan (Fig. 1a). Most studies (87%) sampled bird communities during the breeding season, while 11% included both breeding and overwintering seasons, and 2% focused on fall migration season (Fig. 1d). Finally, 52% of studies took place in forest-dominated landscapes, 26% in mixed forest-open landscapes (i.e. majority of forests), 2% in forest-urban landscapes, 17% in open areas-forest landscapes, and 2% in predominantly open landscapes (Fig. 1e).

Data extraction resulted in 1272 single results (i.e. tested relationships; Fig. 2). Most of these relations were non-significant (63%), while humped-shaped relationships were rarely examined (0.4%). Taxonomic diversity was the most frequently tested response variable (49.6%), followed by abundance (21.5%), functional diversity (13.8%), phylogenetic diversity (10.8%), and species composition (assemblage; 3.8%). Abundance-based metrics showed the highest proportion of significant results, followed by taxonomic diversity, whereas functional diversity, phylogenetic diversity, and species composition mostly resulted in non-significant relationships.

The use of different biological groups (traits) also influenced the ability to detect significant effects of landscape heterogeneity. Relationships using the overall community as response variables (38% of significant effects), species subgroups based on habitat preference (59%), diet/foraging (58%), and nesting location (32%) often led to significant results. In contrast, migration trait (resident vs migrant) usually led to non-significant results (only 11% of significant effects). Protected status, morphological traits, and diet/foraging were the least studied biological groups (5% of single results).

The spatial level of biodiversity measurements was an important factor influencing the result. (Local) alpha-diversity largely dominated the studies (78% of articles, 85% of single results), and often led to non-significant results (29% of significant results; Fig. 1c, 2). In contrast, (landscape-level) gamma-diversity was a lot less common (22% of articles, 15% of single results), but regularly led to significant relationships (82% of significant results; Fig. 1c, 2).

The tested relationships differed according to how the landscape structure was described. Metrics of landscape composition were more than twice more common than configuration ones. Some landscape metrics combined both composition and configurational aspects (mixed), while some studies also included metrics describing the local scale. All metric types led to significant results, although configuration and mixed metrics were more often non-significant than average. Most articles studied a single landscape scale (67%), with an average tested scale of 3313 m (1116 m for alpha-diversity, 11 901 for gamma-diversity; Fig. 3b) and a large range of scales SD = 5400 m. Only 15 studies tested multiple scales, among those the range of landscape scales that led to significant results varied greatly from 100 to 5000 m (overall average = 1018 m; 1104 m for 'All spp.'; 741 m for 'Forest spp.'; Fig. 3a). Moreover, most of found scales of effects were at the extreme scales of the observed ranges (82%; 45% at the smallest scale, 37% at the largest scale). Overall, including every significant relationship we extracted from both multi-scale and single-scale studies, results varied from 100 to 19 700 m (average 5596 m; 1116 m for alpha-diversity, 11 901 m for gamma-diversity; Fig. 3b).

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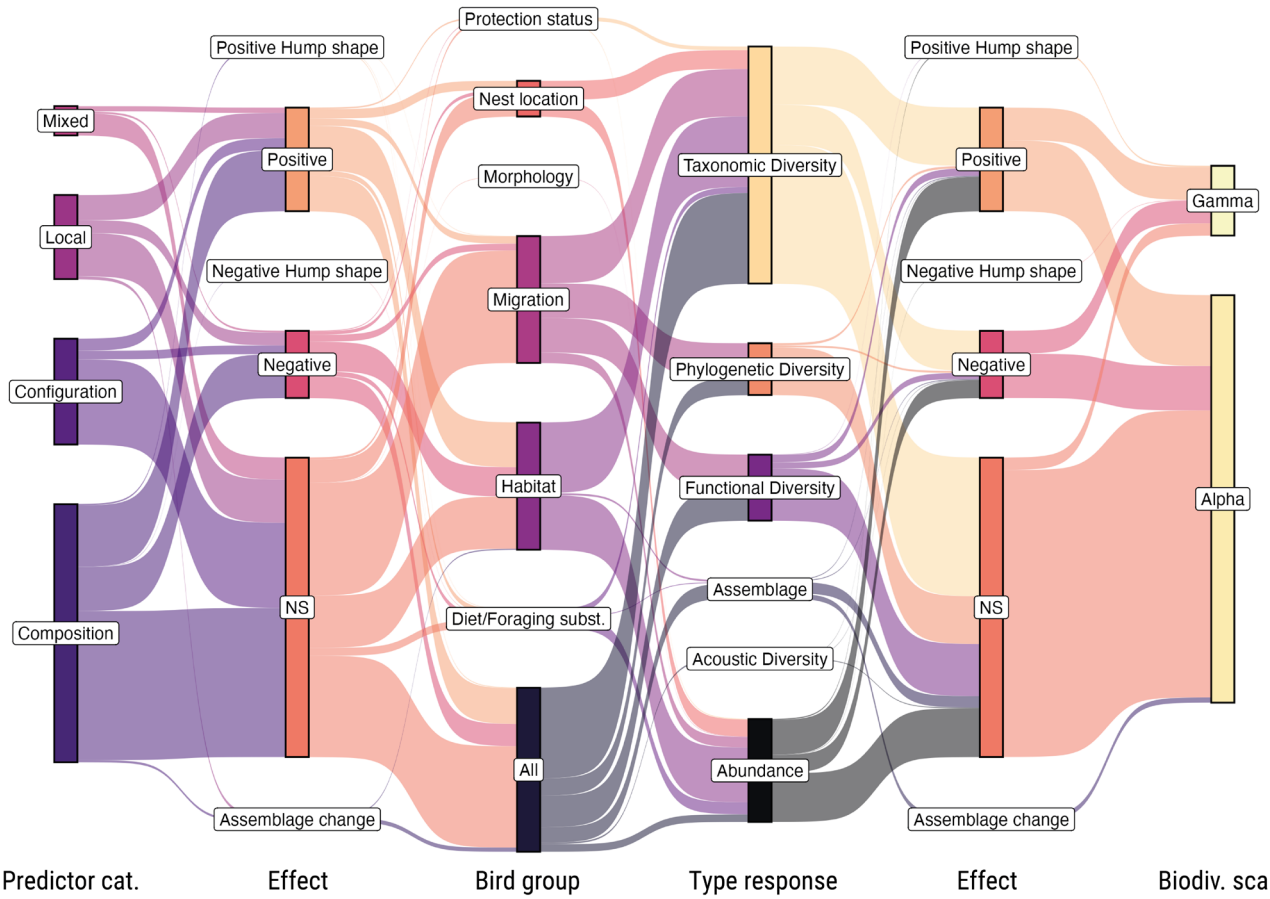

Figure 2. Sankey plot representing the distribution of single tested relationships (n = 1272) across landscape metric categories, significance of detected effects, biological groups (traits), type of biodiversity response variable, and scale of biodiversity measurements. Data on significance of detected effects are repeated twice for better visualization of its link with other factors.

Landscape metric effects on bird communities

We found that most of the significant results testing the effects of landscape metrics on biodiversity were measured at the gamma-scale (Fig. 4, Supporting information). We also found that most of tested types of landscape metrics was related to the composition component of heterogeneity (54% of relationships, 74% of tested landscape metrics; Fig. 4). Among the composition landscape metrics, we found that agriculture amount had usually negative effects on all species, while the composition of all types of habitats (mostly land cover diversity), forest, and, to a lesser extent open areas, had usually positive effects (Fig. 4a, Supporting information). Regarding forest species, we found that the amount, composition and configuration of forest (mostly patch density; Supporting information) had usually positive effects (Fig. 4b). The same landscape metrics often had antagonist negative effects on open area species (Fig. 4c). The open area species were generally positively affected by the amount of agriculture and shrubland, and the open areas composition and configuration (Fig. 4c). We also found that forest composition had usually positive effects on open areas species measured at alpha-diversity scale (Fig. 4c; for more details see the Supporting information). Local environmental variables (especially vegetation structure and land management) also had significant effects for all, forest and open area species at the alpha-diversity scale (Fig. 4).

Discussion

Systematic review

A first striking result from our review is that gamma-diversity, although measured in fewer than 20% of the studies, was much more prone to reveal significant landscape effects on biodiversity than alpha-diversity. Indeed, the scale of biodiversity measurement was among the best predictors discriminating non-significant from significant relationships between landscape and biodiversity metrics (Supporting information). This issue has already been highlighted in discussions on how to interpret landscape effects on biodiversity (Valente et al. 2023). While alpha-diversity is expected to mostly respond to local habitat structure and landscape habitat amount (Fahrig 2013), gamma-diversity is more sensitive to landscape heterogeneity – such as overall heterogeneity, composition, and configuration – because it reflects the full landscape-level species pool (Valente et al. 2023). In contrast, beta-diversity was

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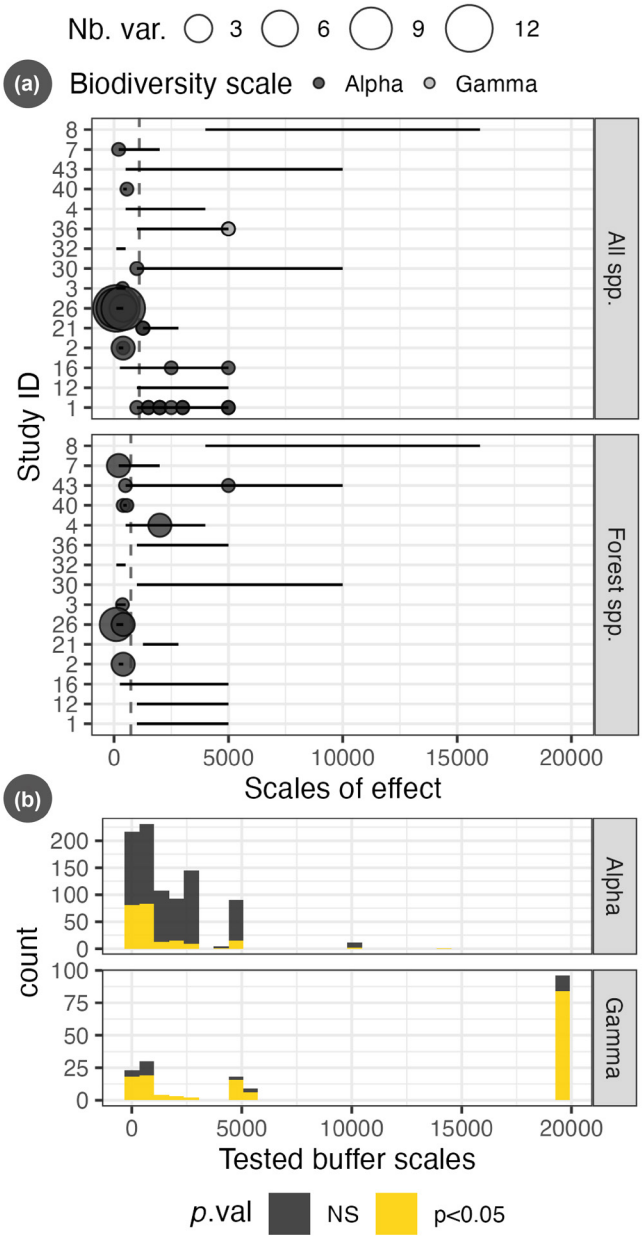

Figure 3. (a) Identified scale of effect in a subset of 13 studies using multiple landscape scales; point show the most relevant scale of significant relationship between a pair of landscape and biodiversity metric for 'All' and 'Forest species'. Dashed lines show average scale of effect ('All spp.' = 1104 m, 'Forest spp.' = 741 m). (b) Count of the non-significant and significant relationship for the different tested scales including all single-scale articles, for the alpha- and gamma-scale of biodiversity measurement. 'Nb. Var.' = number variables.

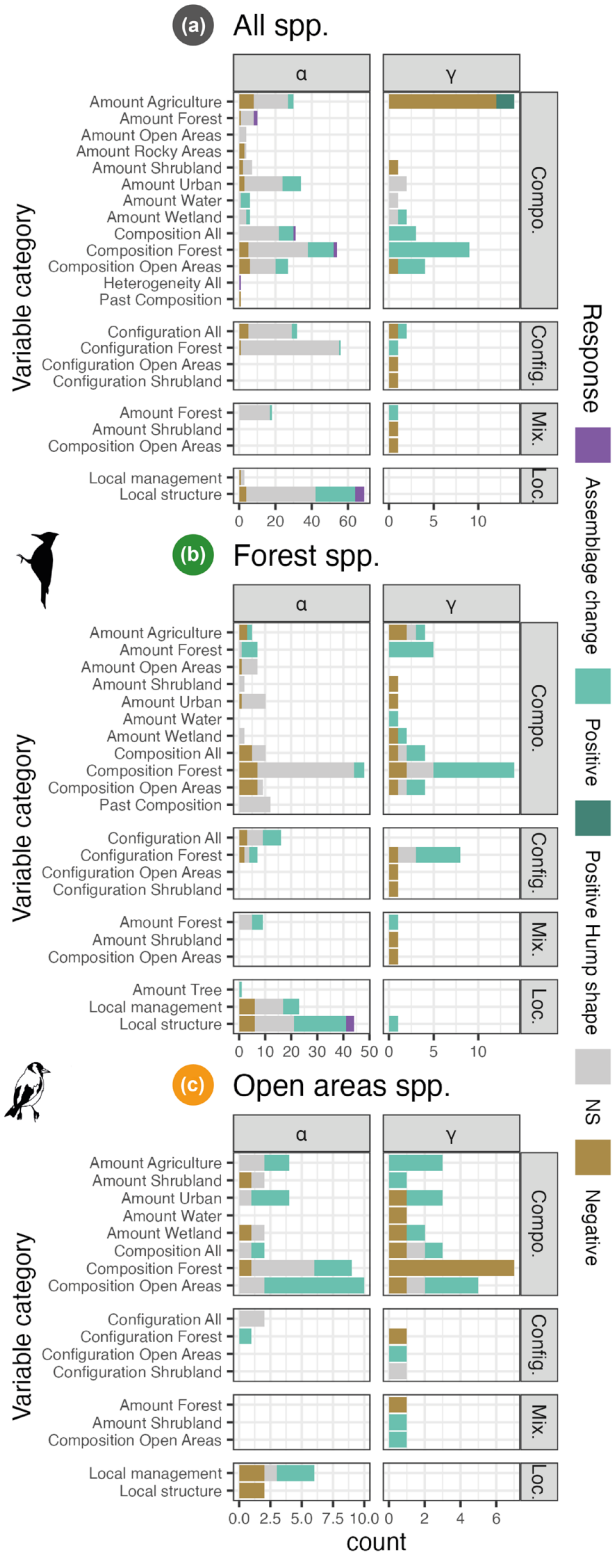

Figure 4. Bar plots counting the number of tested relationships between different types of landscape and local metrics and biodiversity measures for (a) 'All', (b) 'Forest', and (c) 'Open areas species' communities; based on the scale of biodiversity measurement (alpha vs gamma). The x-axis scale is different between panels and species groups.

not assessed in any of the reviewed studies but is expected to be the dominant factor contributing to higher gamma-diversity (Tscharntke et al. 2012).

One objective of our review was to estimate the landscape scale of effect for forest bird communities. The spatial scale at which landscape effects were studied varied widely, with

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most studies (67.5%) testing only a single scale. Among those studies examining multiple scales, the identified scales of effect varied widely depending on the response variable, but were most often found at either the smallest or the largest scale considered. Notably, 28% of studies described landscape heterogeneity at a maximum scale of 500 m, raising concerns about their ability to detect broader-scale effects that may be more relevant for bird communities. As previously noted by Jackson and Fahrig (2015), it is challenging to derive a consistent scale of effect across the literature (Fig. 3a). Furthermore, aside from migration strategy, we surprisingly did not find any studies accounting for bird dispersal capacity – but see Weeks et al. (2023) that showed dispersal capacity matters at global scale.

We found that most of studies sampled bird biodiversity during the breeding season. However, resident bird species, which remain in the same region year-round, often require different habitats across seasons. For example, certain breeding habitats become unsuitable in winter – broadleaf forests, for instance, lose their capacity to protect from predators once leaves have fallen. Yabuhara et al. (2019) showed that specialist bird species were most abundant in coniferous forests during winter, although they left these habitats during the breeding season. Similarly, old-growth forests provide critical resources in winter, such as food (e.g. saproxylic larvae), and shelters (e.g. tree cavities; Jansson and Andrén 2003). Likewise, ecotones, such as forest edges, also support resident birds in winter by fostering fruit-bearing plants (Ciach and Mrowiec 2013). These examples highlight the need to better acknowledge habitat seasonal complementation – i.e. the use of different habitats across seasons – when studying bird communities (see complementation mechanism below). Additionally, the post-breeding dispersal season (i.e. 'fall migration' in Fig. 1) was also entirely overlooked in the reviewed studies, despite its importance for migratory species (McCabe and Olsen 2015). For instance, highly forested landscape dominated by broadleaf forests have been identified as key stopover sites during the post-breeding season for migratory birds in North America (Nguyen et al. 2024).

Identified landscape mechanisms affecting bird biodiversity

Niche partitioning – In forest landscapes, high compositional heterogeneity – such as diverse stand structures and tree species composition across forests patches – is expected to promote greater bird diversity (Fig. 5; a.k.a. the habitat diversity hypothesis sensu Duflot et al. 2022). Heterogeneous habitats provide a wider range of ecological niches, thereby supporting higher gamma-diversity (Benton et al. 2003, Tews et al. 2004, Fahrig et al. 2011). For example, Culbert et al. (2013) found that both canopy cover and biomass level, along with their variability, had strong positive effects on overall and forest bird gamma-diversity in the USA. Similarly, Wood et al. (2004) reported that greater forest habitat diversity was associated with increased bird abundance and species richness. Habitat diversity has often been assessed through land cover diversity metrics including non-forest habitats (e.g. Shannon diversity or land cover richness; Tonetti et al. 2023). These non-forest elements frequently enhanced bird gamma-diversity by supporting species associated with open or non-forested environments (see e.g. Barbaro et al. 2005, Culbert et al. 2013, Klingbeil and Willig 2016).

Cross-habitat spillover – The cross-habitat spillover hypothesis predicts that compositional heterogeneity also

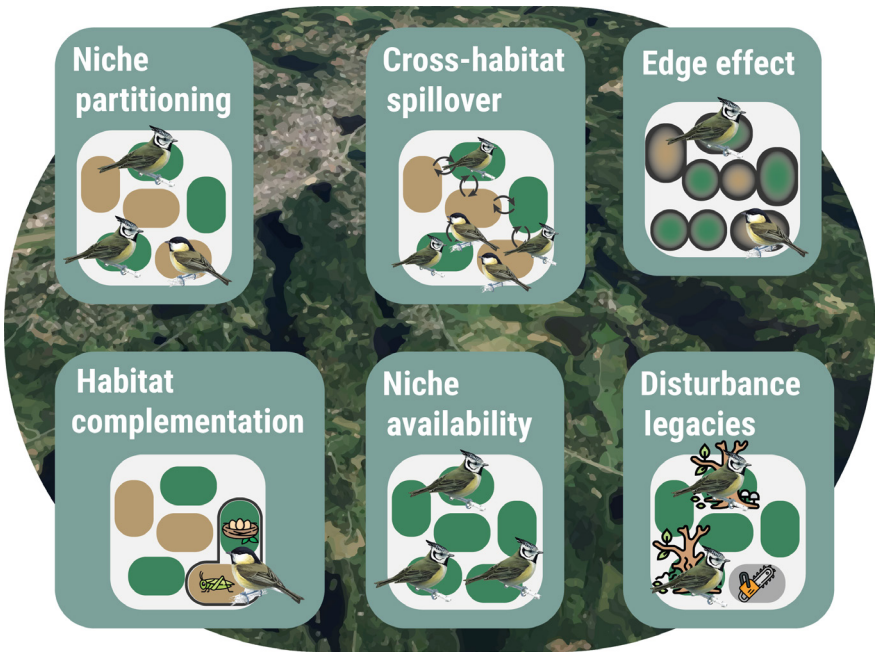

Figure 5. The different identified landscape mechanisms affecting bird communities and their illustrations. Some icons were obtained from 'Flaticon.com'.

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enhances bird diversity at the local (alpha-) scale by facilitating movement of individuals across habitat edges (Fig. 5; Tscharntke et al. 2012, Halstead et al. 2019). This mechanism assumes that species overflow from adjacent habitats, thereby enriching local communities. However, our review revealed mixed support for this hypothesis, with most studies reporting non-significant effects of landscape heterogeneity on local diversity (Fig. 4, Supporting information). In particular, alpha-diversity was most often unrelated to both Shannon diversity of land cover and overall edge density. Nevertheless, some studies partially supported the hypothesis by reporting positive relationships between land cover diversity and local bird species richness or functional diversity (Barbaro et al. 2005, Klingbeil and Willig 2016, Anderle et al. 2022). These findings suggest that landscape-level habitat diversity can, in some cases, contribute to enriching bird communities at local-scale.

Edge effect – Increased configuration heterogeneity connects with the hypothesised negative effects of habitat fragmentation, through reduced patch size, increased isolation, and stronger edge effect (Fig. 5; Villard and Metzger 2014). These factors are generally expected to reduce local (alpha) species diversity (Andrén 1994, Villard et al. 1999, Fahrig 2003). However, this relationship remains debated (see e.g. Valente et al. 2023), as some studies have found that small habitat fragments can support higher densities of species diversity (Riva and Fahrig 2022). For instance, McCabe and Olsen (2015) showed that short-distance migratory birds often concentrated in small patches when surrounding matrix was unhospitable – an example of concentration effect. In contrast, when the matrix is more hospitable, bird densities may decrease locally due to a 'dilution effect' (Tscharntke et al. 2012, Nguyen et al. 2024).

Overall, configuration metrics generally showed little or no effect on bird alpha-diversity (Fig. 4). Most commonly, studies tested contagion (or interspersion) and forest edge density, with results suggesting only limited influence of edge effects on alpha-diversity of all species (Klingbeil and Willig 2016). Nevertheless, some studies mentioned that edge effects might increase nest predation and parasitism risk (Hobson and Bayne 2000, Schmiegelow and Mönkkönen 2002, Klingbeil and Willig 2016). Among the significant results, it remained unclear whether bird diversity responded to habitat configuration per se or to associated changes in habitat composition (i.e. gain or loss of habitats; Schmiegelow and Mönkkönen 2002, Fahrig 2003, Luoto et al. 2004, Barbaro et al. 2005, 2022). The limited impact of configuration might be due to the relatively large proportion of habitat in studied landscapes (i.e. forest), as fragmentation effects are generally thought to emerge when habitat falls below a threshold (i.e. 20–30%; Andrén 1994, Drapeau et al. 2000, Fahrig 2003). Finally, we also observed that habitat configuration is less frequently studied than composition, with an average of 74% of tested landscape metrics per study focusing on compositional aspects.

Habitat complementation – Landscape heterogeneity, in both composition and configuration, can support species that depend on multiple habitat types to meet their ecological needs – a concept known as habitat complementation (Fig. 5; Dunning et al. 1992). For example, Barbaro et al. (2007) found that in temperate forests, species that nest in woodlands and forage in nearby grasslands benefited from the combination of both habitat types. Similarly, in Finnish boreal landscapes, Virkkala et al. (2004) showed that mosaics of semi-natural grasslands and broadleaf forests supported both forest and farmland bird species. Similarly, forest edges can support bird communities within adjacent grasslands (Duflot et al. 2018). Habitat complementation is also important across seasons: Yabuhara et al. (2019) observed that specialist forest birds preferred coniferous forests in winter, while they avoided them during the breeding season.

Niche availability – The habitat amount and thus the niche availability in a landscape – regardless of its spatial configuration – has been recognized as a key driver of bird communities at both alpha- and gamma-levels (Fig. 5; Fahrig 2003, 2013). In our review, forest area generally had positive effects on both alpha- and gamma-diversity of forest bird species (Fig. 4, Supporting information). For example, Radford and Bennett (2007) found that forest extent was the strongest predictor of forest bird gamma-diversity in an agriculture-dominated landscape, outperforming other landscape composition and configuration metrics. However, alpha-diversity of forest bird species often showed non-significant responses to forest composition metrics such as the proportion of coniferous or broadleaf forests (Fig. 4, Supporting information). This may reflect a mismatch between the selected habitat descriptors (e.g. tree species composition) and the actual habitat requirements of the studied bird communities (Fahrig 2013). Supporting this, Mag and Ódor (2015) reported weak or non-significant effects of forest composition metrics such as oak forest proportion, while structural local metrics such as tree diameter and deadwood volume had strong impacts on bird communities. These findings suggest that habitat amount, when measured through structural attributes (direct resources), may be more ecologically meaningful than broad tree species composition (Drapeau et al. 2000). Nevertheless, in conifer-dominated forest, increasing the broadleaf proportion often enhanced bird diversity and abundance (Barbaro et al. 2005, Yabuhara et al. 2019, Felton et al. 2021).

Therefore, not all habitats support bird species equally, and habitat identity emerges as another crucial factor. For instance, the structural complexity of old-growth forests is well known to support many forest specialists such as cavity-nesting birds (Jokimäki and Huhta 1996, Drapeau et al. 2000, Jansson and Andrén 2003, MacKay et al. 2014, Felton et al. 2021). However, in landscapes dominated by structurally similar mature forests, these areas can temporarily compensate for the scarcity of old-growth forest habitats (Schmiegelow and Mönkkönen 2002, Yamaura et al. 2006). In contrast, agricultural areas generally have negative effects on both overall and forest bird gamma-diversity (see e.g. Culbert et al. 2013, Endenburg et al. 2019, Morissette et al. 2019, Wilson et al. 2020), though they support higher diversity of open-area

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bird species. In forest-rich landscapes, small agricultural patches and forest-farmland edges often provide key habitats for these open-area species (Virkkala et al. 2004, Piha et al. 2007, Culbert et al. 2013, Wilson et al. 2020), whereas large cropland fields typically support low avian species richness (Luoto et al. 2004, Wilson et al. 2020).

Disturbance legacies – Several studies have emphasized the impact of landscape disturbances on bird communities. Natural disturbances are often associated with increased landscape heterogeneity, which can enhance bird gamma-diversity (Pickett and White 1985, Hirao et al. 2008, Latif et al. 2020). However, large-scale stand replacing events, such as wildfire, can have substantial negative effects, particularly when they occur at high intensity or frequency (Lindenmayer et al. 2019). In contrast, human-induced disturbances – primarily clear-cutting – can, under certain conditions, support overall gamma-diversity (Drapeau et al. 2000, MacKay et al. 2014), but tend to negatively impact forest-specialist species, especially when replacing old-growth forests at large scale (Fig. 5; Edenius and Elmberg 1996, Drapeau et al. 2000, Jansson and Andrén 2003, Lindenmayer et al. 2019). Clear-cutting may also drive short-term bird movements, leading to a temporary concentration of individuals in remaining patches of mature or old-growth forests, thereby increasing local diversity (Drapeau et al. 2000, Lindenmayer et al. 2019). Importantly, bird communities in logged areas often differ from those in naturally disturbed patches, likely due to differences in disturbance legacies – such as the presence of standing dead trees (snags) and different landscape configurations (Drapeau et al. 2000, Zhao et al. 2013, Lindenmayer et al. 2019).

Perspectives

Despite advances in remote sensing and aerial mapping technologies, forest heterogeneity remains underexplored in landscape-scale biodiversity studies. Forest ecosystems are still too often treated as uniform habitats or simplified into discrete patches defined by a single attribute (e.g. dominant tree species or stand age). Emerging initiatives, such as the 'ForestPaths' project in Europe, which aim to provide detailed maps of forest structure and composition, offer promising tools to better capture the true complexity of forest landscapes. These types of efforts can help identify and differentiate the ecological roles of various forms of old-growth forests that coexist and interact within a landscape (Spies and Franklin 1996).

Despite the valuable body of work reviewed here, substantial knowledge gaps remain regarding the mechanisms governing forest bird responses to landscape structures. For example, the relative importance and potential interaction between habitat composition (including habitat amount) and configuration remain unclear. Moreover, the strong bias toward breeding season studies limits our understanding of bird responses across other critical life-history stages. Seasonal landscape requirements for resident birds in winter, as well as for migrants during pre- and post-breeding periods, are still poorly documented. The increasing availability of high-resolution spatial data, such as LiDAR-based metrics, holds considerable promise for disentangling the ecological processes underpinning bird-landscape relationships. A clearer grasp of these mechanisms, along with better identification of scales of effect, is essential to develop targeted and effective conservation strategies for forest bird communities.

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Author contributions

Jérémy Cours: Conceptualization (equal); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (equal); Visualization (lead); Writing - original draft (lead). **Rémi Duflot**: Conceptualization (equal); Funding acquisition (lead); Methodology (equal); Project administration (lead); Supervision (lead); Validation (lead); Writing - review & editing (lead).

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Supporting information

The Supporting information associated with this article is available with the online version.

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